UKF – Unscented Kalman filter algorithm

Non-linear system

Process $\quad \mathbf{x_{k+1}} = \mathbf{f}(\mathbf{x_k}, \mathbf{u_k}) + \mathbf{w_k}, \qquad \mathbf{w_k} \sim (\mathbf{0}, \mathbf{Q_k})$

Measurement $\quad \mathbf{y_k} = \mathbf{h}(\mathbf{x_k}) + \mathbf{v_k}, \qquad \mathbf{v_k} \sim (\mathbf{0}, \mathbf{R_k})$

Predict $\mathbf{k-1 \rightarrow k}$ (time update)

Given $\hat{\mathbf{x}}_{\mathbf{k-1}}^{+}, \ \mathbf{P}_{\mathbf{k-1}}^{+}$

UT(f):

Sigma-points

$$\hat{\mathbf{x}}_{\mathbf{k-1}}^{(\mathbf{i})} = \hat{\mathbf{x}}_{\mathbf{k-1}}^{+} + \tilde{\mathbf{x}}^{(\mathbf{i})}, \qquad \mathrm{i} = 1, \dots, 2\mathrm{n}$$

$$\tilde{\mathbf{x}}^{(\mathbf{i})} = \left(\sqrt{\mathrm{n}\, \mathbf{P}_{\mathbf{k-1}}^{+}}\right)_{\mathrm{i'th\ col.}}, \qquad \mathrm{i} = 1, \dots, \mathrm{n}$$

$$\tilde{\mathbf{x}}^{(\mathbf{n+i})} = -\left(\sqrt{\mathrm{n}\, \mathbf{P}_{\mathbf{k-1}}^{+}}\right)_{\mathrm{i'th\ col.}}, \qquad \mathrm{i} = 1, \dots, \mathrm{n}$$

Propagate

$$\hat{\mathbf{x}}_{\mathbf{k}}^{(\mathbf{i})} = \mathbf{f}(\hat{\mathbf{x}}_{\mathbf{k-1}}^{(\mathbf{i})}, \mathbf{u_k}), \qquad \mathrm{i} = 1, \dots, 2\mathrm{n}$$

Weights

$$\mathrm{w}^{(\mathrm{i})} = \frac{1}{2\mathrm{n}}, \qquad \mathrm{i} = 1, \dots, 2\mathrm{n}$$

A priori-estimates

$$\hat{\mathbf{x}}_{\mathbf{k}}^{-} = \sum_{\mathrm{i}=1}^{2\mathrm{n}} \mathrm{w}^{(\mathrm{i})}\, \hat{\mathbf{x}}_{\mathbf{k}}^{(\mathbf{i})}$$

$$\mathbf{P}_{\mathbf{k}}^{-} = \sum_{\mathrm{i}=1}^{2\mathrm{n}} \mathrm{w}^{(\mathrm{i})}\, (\hat{\mathbf{x}}_{\mathbf{k}}^{(\mathbf{i})} - \hat{\mathbf{x}}_{\mathbf{k}}^{-})\, (\ )^{\mathbf{T}} + \mathbf{Q_{k-1}}$$

Update $\mathbf{k^- \rightarrow k^+}$ (measurement update)

Given $\hat{\mathbf{x}}_{\mathbf{k}}^{-}, \ \mathbf{P}_{\mathbf{k}}^{-}$

UT(h):

Sigma-points

$$\hat{\mathbf{x}}_{\mathbf{k}}^{(\mathbf{i})} = \hat{\mathbf{x}}_{\mathbf{k}}^{-} + \tilde{\mathbf{x}}^{(\mathbf{i})}, \qquad \mathrm{i} = 1, \dots, 2\mathrm{n}$$

$$\tilde{\mathbf{x}}^{(\mathbf{i})} = \left(\sqrt{\mathrm{n}\, \mathbf{P}_{\mathbf{k}}^{-}}\right)_{\mathrm{i'th\ col.}}, \qquad \mathrm{i} = 1, \dots, \mathrm{n}$$

$$\tilde{\mathbf{x}}^{(\mathbf{n+i})} = -\left(\sqrt{\mathrm{n}\, \mathbf{P}_{\mathbf{k}}^{-}}\right)_{\mathrm{i'th\ col.}}, \qquad \mathrm{i} = 1, \dots, \mathrm{n}$$

Propagate

$$\hat{\mathbf{y}}_{\mathbf{k}}^{(\mathbf{i})} = \mathbf{h}(\hat{\mathbf{x}}_{\mathbf{k}}^{(\mathbf{i})}), \qquad \mathrm{i} = 1, \dots, 2\mathrm{n}$$

Weights

$$\mathrm{w}^{(\mathrm{i})} = \frac{1}{2\mathrm{n}}, \qquad \mathrm{i} = 1, \dots, 2\mathrm{n}$$

A priori-estimates

$$\hat{\mathbf{y}}_{\mathbf{k}} = \sum_{\mathrm{i}=1}^{2\mathrm{n}} \mathrm{w}^{(\mathrm{i})}\, \hat{\mathbf{y}}_{\mathbf{k}}^{(\mathbf{i})}$$

$$\mathbf{P_y} = \sum_{\mathrm{i}=1}^{2\mathrm{n}} \mathrm{w}^{(\mathrm{i})}\, (\hat{\mathbf{y}}_{\mathbf{k}}^{(\mathbf{i})} - \hat{\mathbf{y}}_{\mathbf{k}})\, (\ )^{\mathbf{T}} + \mathbf{R_k}$$

$$\mathbf{P_{xy}} = \sum_{\mathrm{i}=1}^{2\mathrm{n}} \mathrm{w}^{(\mathrm{i})} \left(\hat{\mathbf{x}}_{\mathbf{k}}^{(\mathbf{i})} - \hat{\mathbf{x}}_{\mathbf{k}}^{-}\right) \left(\hat{\mathbf{y}}_{\mathbf{k}}^{(\mathbf{i})} - \hat{\mathbf{y}}_{\mathbf{k}}\right)^{\mathbf{T}}$$

New observation $\quad \mathbf{y_k}$

Kalman gain $\quad \mathbf{K_k} = \mathbf{P_{xy}}\, \mathbf{P_y}^{-1}$ **(see note)**

A posteriori-estimates

$$\hat{\mathbf{x}}_{\mathbf{k}}^{+} = \hat{\mathbf{x}}_{\mathbf{k}}^{-} + \mathbf{K_k}\, (\mathbf{y_k} - \hat{\mathbf{y}}_{\mathbf{k}})$$

$$\mathbf{P}_{\mathbf{k}}^{+} = \mathbf{P}_{\mathbf{k}}^{-} - \mathbf{K_k}\, \mathbf{P_y}\, \mathbf{K_k}^{\mathbf{T}} \qquad \textbf{(see note)}$$

Initialization

$$\hat{\mathbf{x}}_{\mathbf{0}}^{+} = \mathrm{E}[\mathbf{x_0}], \qquad \mathbf{P}_{\mathbf{0}}^{+} = \mathrm{Cov}[\mathbf{x_0}]$$

Note on equations for $\mathbf{K_k}$ and $\mathbf{P_k^+}$

For a linear system with measurement model

$$\mathbf{y_k} = \mathbf{H}\,\mathbf{x_k} + \mathbf{v_k}, \quad \mathbf{v_k} \sim (\mathbf{0}, \mathbf{R_k}),$$

the equation for the Kalman gain is

$$\mathbf{K_k} = \mathbf{P_k^-}\mathbf{H^T}(\mathbf{H}\mathbf{P_k^-}\mathbf{H^T} + \mathbf{R_k})^{-1}$$

and the posterior estimate of the state-estimation covariance (MSE) is

$$\mathbf{P_k^+} = \mathrm{E}[(\mathbf{x_k} - \hat{\mathbf{x}}_\mathbf{k}^+)(\mathbf{x_k} - \hat{\mathbf{x}}_\mathbf{k}^+)^\mathbf{T}] = (\mathbf{I} - \mathbf{K_k}\mathbf{H})\mathbf{P_k^-} = \mathbf{P_k^-} - \mathbf{K_k}\mathbf{H}\mathbf{P_k^-}$$

These equations cannot be used for the UKF since for a non-linear system there is no $\mathbf{H}$ matrix (but instead the non-linear function $\mathbf{h}$).

However, an alternative form for the equations can be found by noticing that – given the above linear measurement model - the covariance matrix for estimated observations is

$$\mathbf{P_y} = \mathrm{E}[(\mathbf{y_k} - \hat{\mathbf{y}}_\mathbf{k})(\mathbf{y_k} - \hat{\mathbf{y}}_\mathbf{k})^\mathbf{T}] = \mathbf{H}\mathbf{P_k^-}\mathbf{H^T} + \mathbf{R_k}$$

where

$$\mathbf{P_k^-} = \mathrm{E}[(\mathbf{x_k} - \hat{\mathbf{x}}_\mathbf{k}^-)(\mathbf{x_k} - \hat{\mathbf{x}}_\mathbf{k}^-)^\mathbf{T}]$$

Moreover, the cross-covariance matrix between observation and state estimates is (again considering the linear measurement model)

$$\mathbf{P_{xy}} = \mathrm{E}[(\mathbf{x_k} - \hat{\mathbf{x}}_\mathbf{k}^-)(\mathbf{y_k} - \hat{\mathbf{y}}_\mathbf{k})^\mathbf{T}] = \mathbf{P_k^-}\mathbf{H^T}$$

Therefore the Kalman gain equation for the linear system and the standard Kalman filter could also be written as

$$\mathbf{K_k} = \mathbf{P_k^-}\mathbf{H^T}(\mathbf{H}\mathbf{P_k^-}\mathbf{H^T} + \mathbf{R_k})^{-1} = \mathbf{P_{xy}}\,\mathbf{P_y^{-1}}$$

and the equation for the posterior estimate of the state-estimation covariance (MSE) could also be written as

$$\mathbf{P_k^+} = \mathbf{P_k^-} - \mathbf{K_k}\mathbf{H}\mathbf{P_k^-} = \mathbf{P_k^-} - \mathbf{K_k}\mathbf{P_y}\mathbf{K_k^T}$$

This form of the equations for Kalman gain and MSE estimate can be used in the UKF algorithm since they only involve estimated covariance and cross-covariance matrices and these are estimated using the sigma-points as explained above.

UKF (vs. EKF) advantages

- No linearization of the non-linear functions $\mathbf{f}$ and $\mathbf{h}$
- Sigma points are propagated exact through the non-linear functions (no approximations)
- No Jacobian matrices are required (no differentiation)
- UKF estimates state and MSE (first and second order moments) correctly up to third order in Taylor series terms – EKF only up to first order
- UKF algorithm complexity comparable with EKF

Case of non-additive noise

If the process and measurement noise is not additive, i.e.

$$\mathbf{x_{k+1}} = \mathbf{f}(\mathbf{x_k}, \mathbf{u_k}, \mathbf{w_k})$$
$$\mathbf{y_k} = \mathbf{h}(\mathbf{x_k}, \mathbf{v_k}),$$

the UKF sigma-point approach can still be used by augmenting the state vector with the process and measurement noise i.e.

$$\mathbf{x_{aug,k}} = \begin{bmatrix} \mathbf{x_k} \\ \mathbf{w_k} \\ \mathbf{v_k} \end{bmatrix}$$

This means that the state (mean) and covariance of the noise states are also estimated by the sigma-points a priori and a posteriori estimates. The equations for the above UKF algorithm will then still be the same except that $\mathbf{Q_{k-1}}$ and $\mathbf{R_k}$ should not be added to the state covariance estimates (since in this case we are estimating the noise covariances).

The algorithm can in this case be initialized as

$$\hat{\mathbf{x}}^{+}_{\mathbf{aug,0}} = \begin{bmatrix} \mathrm{E}[\mathbf{x_0}] \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

$$\mathbf{P}^{+}_{\mathbf{aug,0}} = \begin{bmatrix} \mathrm{Cov}[\mathbf{x_0}] & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{Q_0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{R_0} \end{bmatrix}$$

Generalized UT/UKF

If there is available information about the noise distributions (e.g., Gaussian), the UKF performance can be improved slightly by using a generalized form of the UT presented somewhat compactly below (where κ is a user-chosen free parameter):

- $2\mathrm{n} + 1$ sigma-points
- $\mathbf{x}^{(\mathbf{0})} = \bar{\mathbf{x}}$
- $\mathbf{x}^{(\mathbf{i})} = \bar{\mathbf{x}} + \tilde{\mathbf{x}}^{(\mathbf{i})}, \quad \mathrm{i} = 1, \ldots, 2\mathrm{n}$
- $\tilde{\mathbf{x}}^{(\mathbf{i})} = \pm \left(\sqrt{(\mathrm{n} + \kappa)\, \mathbf{P}} \right)_{\mathrm{i'th\ col.}}$
- $\mathrm{w}^{(0)} = \frac{\kappa}{\mathrm{n}+\kappa}$
- $\mathrm{w}^{(\mathrm{i})} = \frac{1}{2(\mathrm{n}+\kappa)}, \quad \mathrm{i} = 1, \ldots, 2\mathrm{n}$
- $\mathbf{y}^{(\mathbf{i})} = \mathbf{f}(\mathbf{x}^{(\mathbf{i})})$
- $\bar{\mathbf{y}}_{\mathbf{u}} = \sum_{\mathbf{i=0}}^{\mathbf{2n}} \mathrm{w}^{(\mathrm{i})} \mathbf{y}^{(\mathbf{i})}$
- $\mathbf{P}_{\mathbf{y,u}} = \sum_{\mathbf{i=0}}^{\mathbf{2n}} \mathrm{w}^{(\mathrm{i})} \left(\mathbf{y}^{(\mathbf{i})} - \bar{\mathbf{y}}_{\mathbf{u}} \right) \left(\mathbf{y}^{(\mathbf{i})} - \bar{\mathbf{y}}_{\mathbf{u}} \right)^{\mathbf{T}}$

For Gaussian noise $\mathrm{n} + \kappa = 3$ is optimal and this form for using sigma-points for estimating states and covariances can the be used in the UKF algorithm at UT(**f**) and UT(**h**).
Note that for $\kappa = 0$ the generalized UT specializes to the one presented previously and used above in the UKF algorithm